A. Official-source conditions

Archived codes by workflow. Each task row has CANN and CUDA columns; G uses ROCm/HIP in the second column.

Task		M1 Find		M2 Fetch		M3 Adequacy		M4 Version	
		CANN	CUDA‡	CANN	CUDA‡	CANN	CUDA‡	CANN	CUDA‡
Environment and installation									
I	Version compatibility	3	3	4	5	4	5	4	4
J	Installation	5	4	4	5	5	5	2	3
K	Container setup	5	5	5	5	5	4	4	5
Operator development									
C	Library selection	4	5	4	5	3	5	3	5
D	Custom operator	2	5	2	5	—	5	2	3
L	Operator accuracy	2	2	4	5	5	5	2	5
M	Dynamic shapes / tiling	2	5	4	5	4	5	4	4
N	Operator fusion	5	2	4	5	5	4	2	5
O	Framework integration	2	5	4	5	5	5	3	3
Training									
F	Distributed training	4	5	4	5	4	5	2	3
P	Mixed precision	3	5	4	5	4	5	4	5
Q	Memory / OOM	5	5	4	5	5	5	4	5
R	Training convergence	3	3	4	5	5	4	3	5
Inference and deployment									
A	Model conversion	4	5	4	5	4	5	2	4
H	Quantization	3	5	4	5	3	5	2	3
S	Inference serving	5	5	4	5	4	5	3	4
T	Dynamic inference	3	3	4	5	5	5	2	5
Performance and optimization									
B	Profiling	4	4	4	5	4	5	2	4
U	Memory / occupancy	4	5	4	5	4	5	5	5
V	Compute-transfer overlap	4	5	4	5	4	5	2	5
W	Multi-device communication	2	5	4	5	3	4	3	4
Debugging									
E	Error-code diagnosis	3	4	4	5	2	5	2	5
X	Memory-error diagnosis	5	5	4	5	5	5	2	3
Migration									
Y	Chip migration	2	5	4	5	4	5	3	4
Z	Programming concepts	5	5	4	5	5	4	5	5
Migration analogy (not in CANN/CUDA summaries)									
G	Migration analogy [CANN / ROCm-	4	5	4	5	4	5	4	3

M1–M10 are archived ordinal codes (1–5; higher is more favorable; M8 denotes lower effort). “—” = unobserved core adequacy.

* M7 is an archived estimate. † M11 is the historical heuristic availability index. ‡ For G, CUDA‡ = ROCm/HIP; it is excluded from CANN/CUDA summaries.